

# PYTHON FULL STACK DEVELOPMENT

Curriculum



# **PYTHON FULL STACK DEVELOPMENT**

## **Why This Course?**

Build complete, real-world web applications with Python, Django, React and MySQL. This course combines frontend, backend, database, REST API and deployment skills, giving learners practical experience in building and integrating full-stack applications.

## **Market Demand**

Python Full Stack Development is highly relevant in today's software market, as Python is widely used for backend development, APIs, automation and modern web applications. Combining Python with Django, React, REST APIs and MySQL enables developers to build complete, scalable applications and prepares learners for a broad range of full-stack software development opportunities.

## Course Highlights

3-Month Industry-Oriented Training

Hands-on Full-Stack Development

React + Python + Django + MySQL

REST API & Full-Stack Integration

Real-World Projects & Capstone Project

Individual Mentor Support

Git, GitHub & Deployment Basics

Career-Focused Practical Learning

## Front-End



HTML



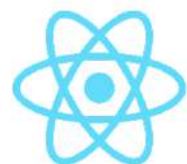
CSS



JS



Bootstrap



React.js

## Back-End



Python



Django

## Database



SQL



My SQL

## Tools & Technologies



Git



My SQL



Postman



REST APIs

## MODULE 1 : HTML

- Introduction to HTML and Web Development
- Structure of an HTML Document
- HTML Elements, Tags and Attributes
- Headings, Paragraphs and Text Formatting
- Links and Navigation

- Images, Audio and Video
- Lists and Tables
- Forms and Form Controls
- Input Types and Form Validation
- Semantic HTML5 Elements
- iframe and Multimedia Elements

## **MODULE 2: CSS & BOOTSTRAP**

- Introduction to CSS
- Inline, Internal and External CSS
- CSS Selectors
- Colors, Fonts and Text Styling
- Backgrounds and Borders
- CSS Box Model
- Display and Positioning
- Flexbox
- CSS Grid
- Pseudo Classes and Pseudo Elements
- Transitions and Animations
- Responsive Web Design
- Media Queries
- Bootstrap Introduction and Setup
- Bootstrap Grid System and Breakpoints
- Bootstrap Typography and Utilities
- Buttons, Forms and Tables

- Navbar, Cards, Alerts and Modals
- Responsive UI Development using Bootstrap

## **MODULE 3: JAVASCRIPT**

- JavaScript Introduction and Fundamentals
- Variables, Data Types and Operators
- Conditional Statements
- Loops
- Functions
- Arrays and Objects
- String and Array Methods
- ES6 Features
- Arrow Functions
- Destructuring
- Spread and Rest Operators
- DOM Manipulation
- Events and Event Handling
- Form Validation
- JSON
- Local Storage and Session Storage
- Error Handling
- Promises
- Async/Await
- Fetch API
- Consuming REST APIs

## **MODULE 4: REACT JS**

- Introduction to React
- React Environment Setup
- React Project Structure
- JSX
- Components
- Props
- State Management
- Event Handling
- Conditional Rendering
- Lists and Keys
- Forms and Form Handling
- React Hooks
  - useState
  - useEffect
  - useRef
  - useContext
- Component Lifecycle Concepts
- React Router
- Navigation and Routing
- API Integration using Fetch/Axios
- CRUD Operations
- Form Validation

- Context API
- Reusable Components
- React Application Development

## **MODULE 5: PYTHON PROGRAMMING**

- Introduction to Python
- Python Installation and Environment Setup
- Variables and Data Types
- Type Casting
- Operators
- Conditional Statements
- Loops and Control Statements
- Strings and String Operations
- Lists, Tuples, Sets and Dictionaries
- Functions and Arguments
- Lambda Functions
- Modules and Packages
- Exception Handling
- File Handling
- JSON Handling
- Virtual Environment and pip
- Working with External Packages

- Object-Oriented Programming
  - Classes and Objects
  - Constructors
  - Inheritance
  - Polymorphism
  - Encapsulation
  - Abstraction

## **MODULE 6: DJANGO**

- Introduction to Django
- Django Installation and Project Setup
- Django Project and Application Structure
- URL Routing
- Views
- Request and Response Handling
- Templates and Template Language
- Static and Media Files
- Django Models
- Django ORM
- Database Configuration
- Migrations
- Model Relationships
- Forms and Model Forms
- CRUD Operations
- Django Admin Panel

- Authentication and Authorization
- Sessions and Cookies
- Middleware
- Validation and Error Handling
- Pagination and Search

## **MODULE 7: REST API WITH DJANGO**

- Introduction to REST API
- REST Architecture
- HTTP Methods
  - GET
  - POST
  - PUT
  - PATCH
  - DELETE
- HTTP Status Codes
- JSON Request and Response
- Django REST Framework
- Serializers
- API Views
- Generic Views
- ViewSets and Routers
- CRUD REST APIs

- Authentication and Permissions
- API Validation and Error Handling
- Postman API Testing
- CORS Basics
- Connecting React with Django REST APIs

## **MODULE 8: SQL & MYSQL**

- Introduction to Databases and RDBMS
- MySQL Installation and Setup
- Database and Table Creation
- Data Types
- DDL, DML and DQL
- SELECT Queries
- INSERT, UPDATE and DELETE
- WHERE, ORDER BY and DISTINCT
- Aggregate Functions
- GROUP BY and HAVING
- Primary Key and Foreign Key
- Constraints
- One-to-One, One-to-Many and Many-to-Many Relationships
- Joins
  - INNER JOIN
  - LEFT JOIN
  - RIGHT JOIN

- Subqueries
- UNION and UNION ALL
- CASE Statements
- Views
- Indexes
- Stored Procedures
- Functions
- Triggers
- Transactions
- COMMIT and ROLLBACK
- Database Normalization
- Real-Time SQL Query Practice

## **MODULE 9: FULL-STACK APPLICATION DEVELOPMENT**

- React Frontend and Django Backend Integration
- Consuming Django REST APIs from React
- CRUD Application Development
- User Registration and Login
- Authentication and Authorization
- Connecting Django with MySQL
- Database Operations using Django ORM
- Frontend Form Integration
- API Error Handling
- Complete Full-Stack Application Development

## **MODULE 10: PROJECT & DEPLOYMENT**

- Git and GitHub Basics
- Project Structure and Version Control
- Environment Variables
- Production Configuration Basics
- Frontend and Backend Deployment Concepts
- Database Backup and Management
- Full-Stack Project Development

## **CAPSTONE PROJECT**

- Develop a Complete Full-Stack Application using:
  - HTML + CSS + Bootstrap + JavaScript + React
  - Python + Django + Django REST Framework
  - MySQL
- Example Projects:
  - E-Commerce Application
  - Employee Management System
  - Learning Management System
  - Job Portal
  - Task Management System

